Part 3
Analyzing System Process Requirements

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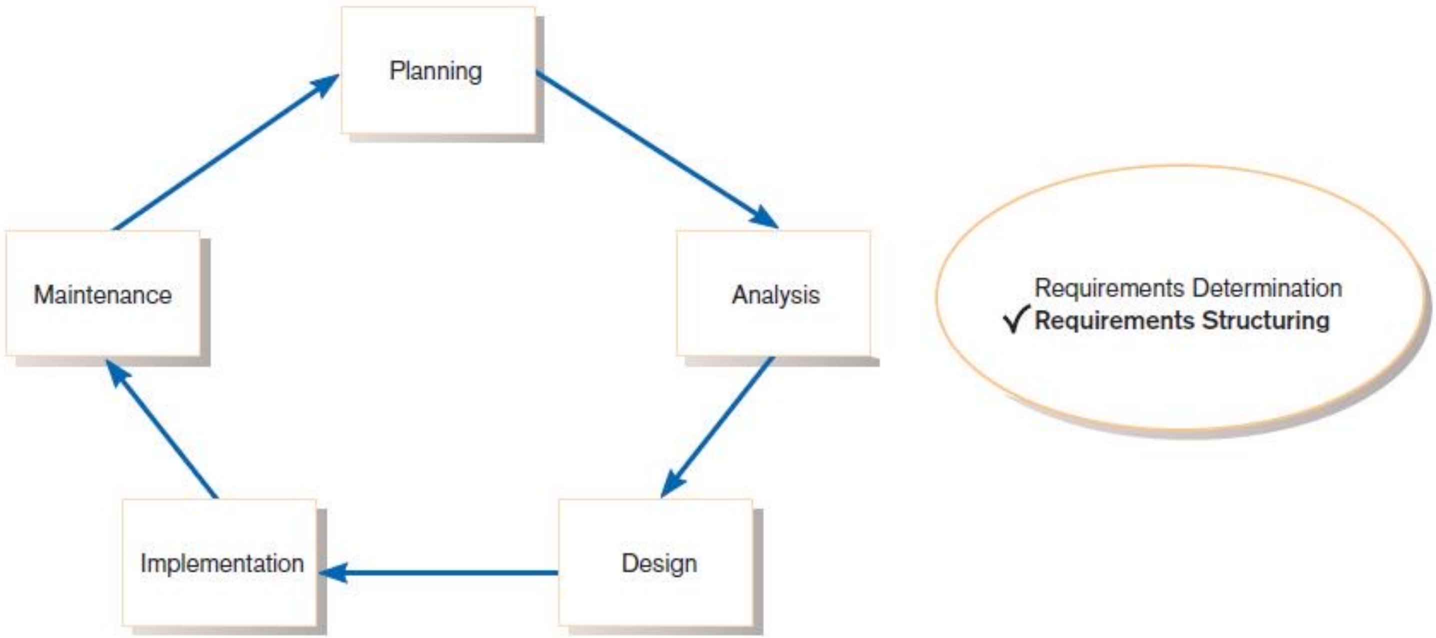
Process Modeling



Systems development life cycle with the analysis phase highlighted

Process Modeling (Cont.)

- Graphically represent the processes that capture, manipulate, store, and distribute data between a system and its environment and among system components.

Process Modeling (Cont.)

- Utilize information gathered during requirements determination.
- Processes and data structures are modeled.

Data Flow Diagram

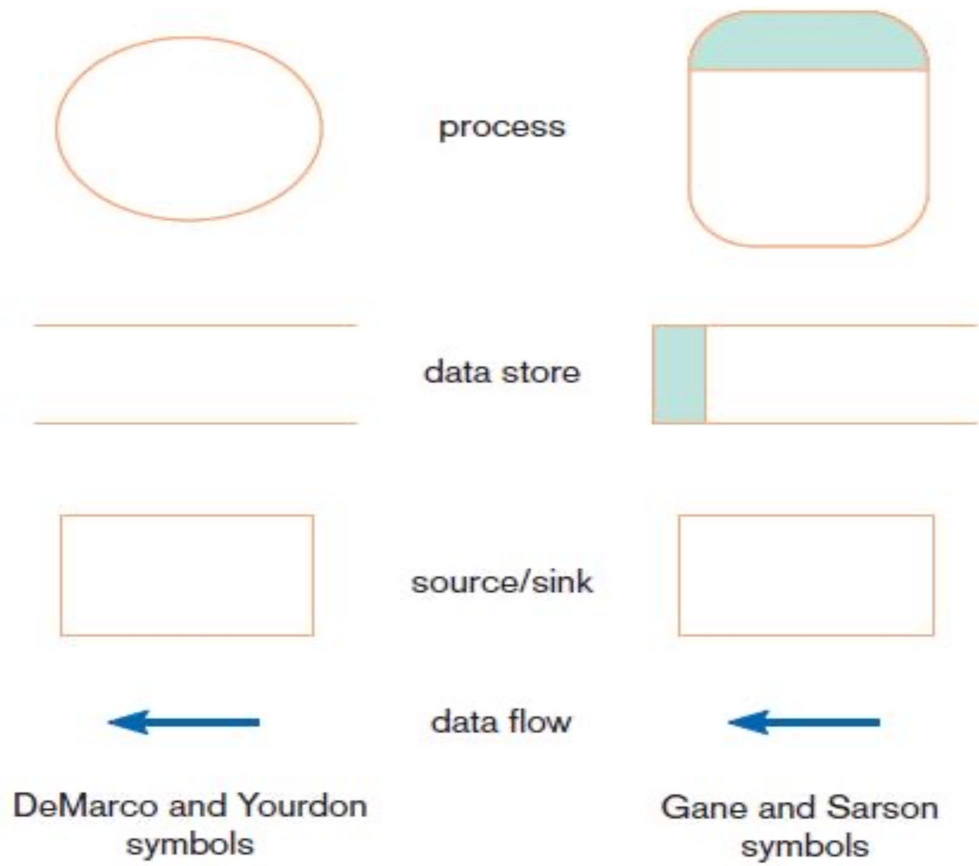
- Represent both physical and logical information systems
- Only four symbols are used

Data Flow Diagram (Cont.)

- Useful for depicting purely logical information flows
- DFDs that detail physical systems differ from system flowcharts which depict details of physical computing equipment

Definitions and Symbols

Comparison of DeMarco and Yourdon and Gane and Sarson DFD symbol sets



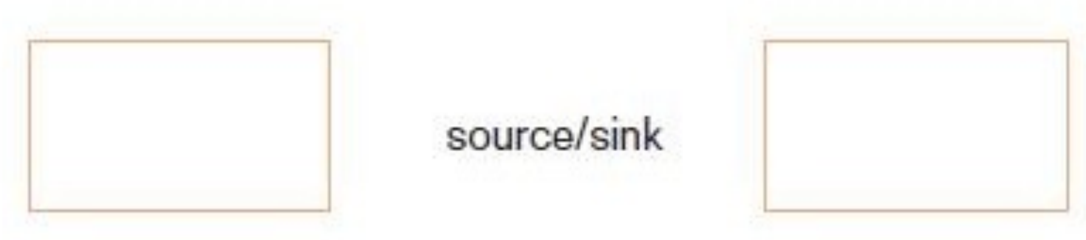
Definitions and Symbols (Cont.)

- **Process**: work or actions performed on data (inside the system)

- **Data store**: data at rest (inside the system)

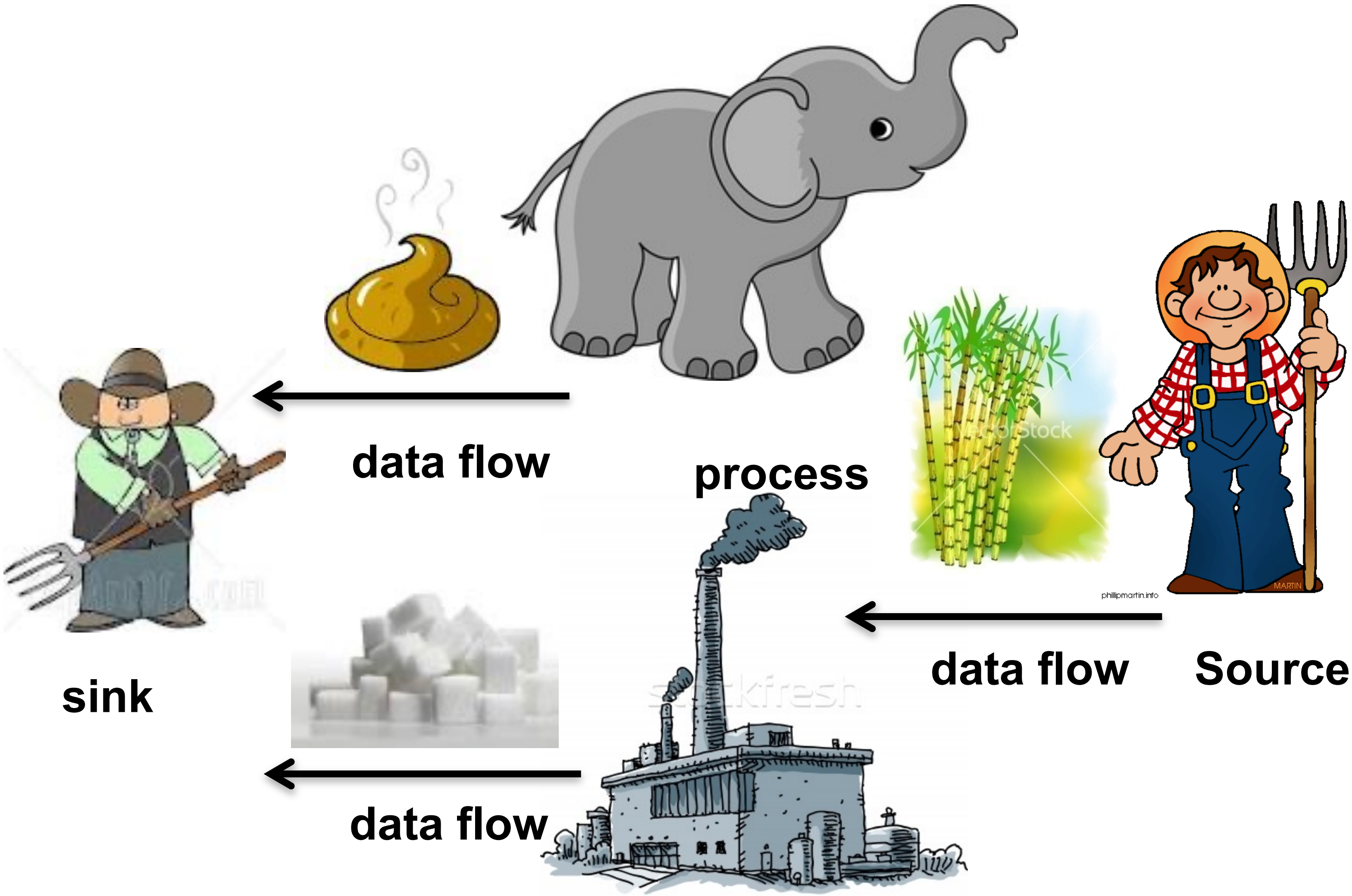
Definitions and Symbols (Cont.)

- **Source/sink**: external entity that is origin or destination of data (outside the system)

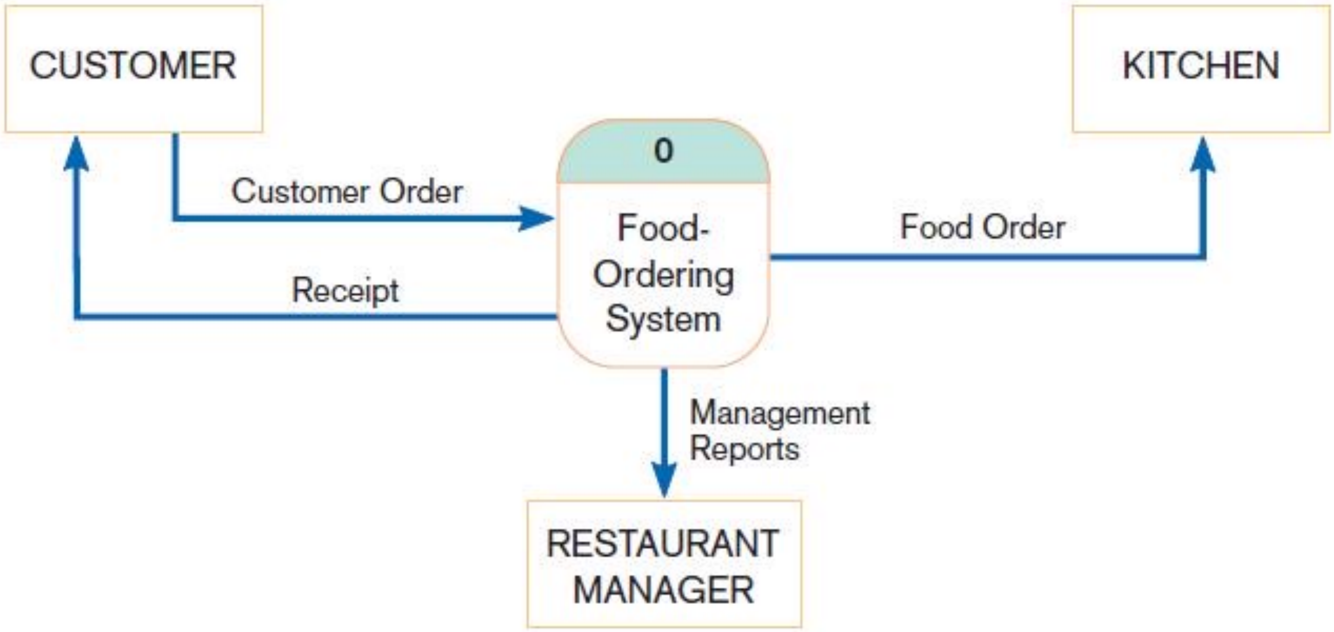


- **Data flow**: arrows depicting movement of data

A Metaphor of Data Flow Diagram



Context Diagram



Context diagram of Hoosier Burger's food-ordering system

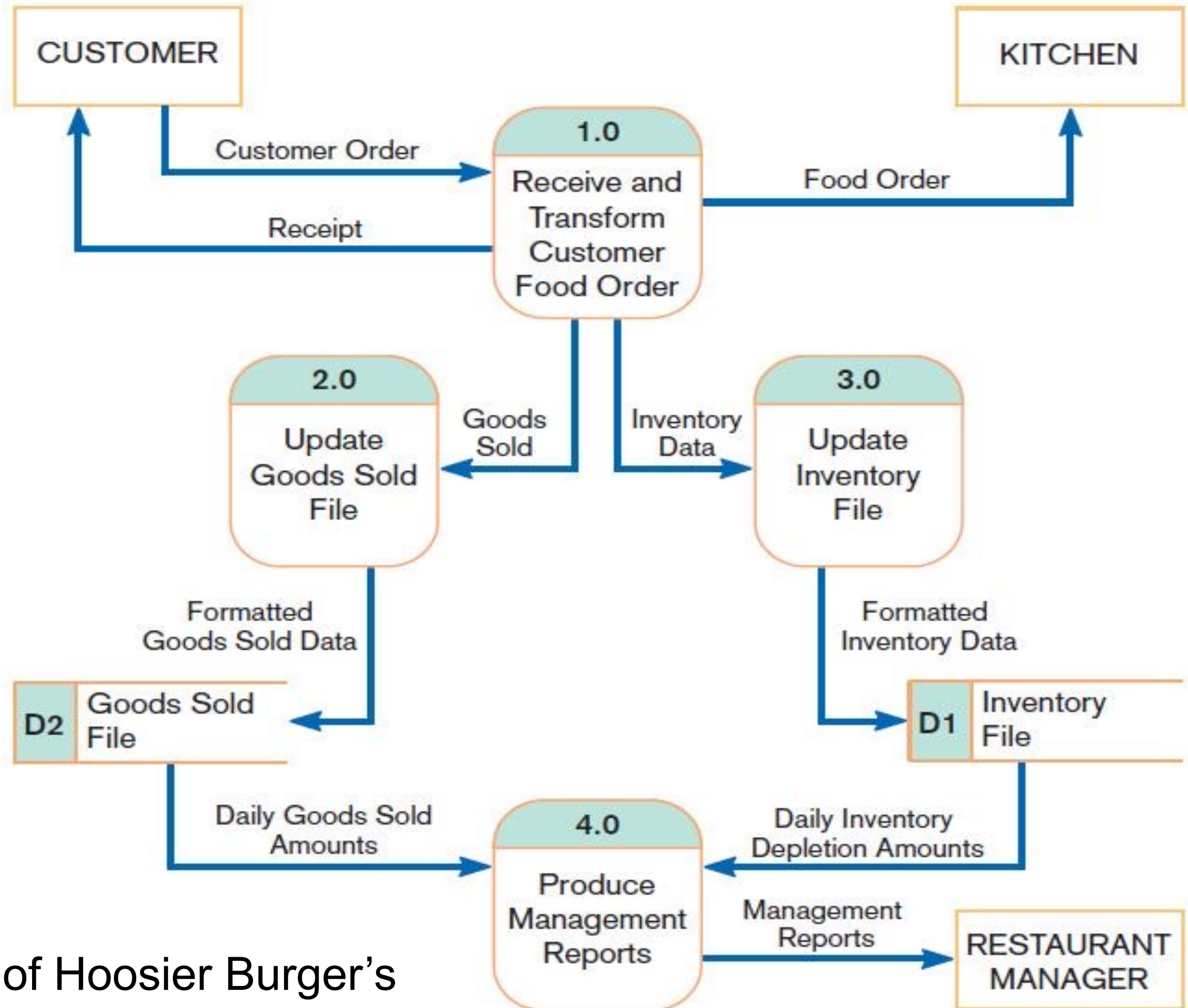
Context diagram

- **Context diagram** is an overview of an organizational system that shows:
  - the system boundaries.
  - external entities that interact with the system.
  - major information flows between the entities and the system.
- Note: only one process symbol, and no data stores shown

Developing DFDs (Cont.)

- **Level-0 diagram** is a data flow diagram that represents a system's major processes, data flows, and data stores at a high level of detail.
  - Processes are labeled 1.0, 2.0, etc. These will be decomposed into more primitive (lower-level) DFDs.

Level-0 Diagram

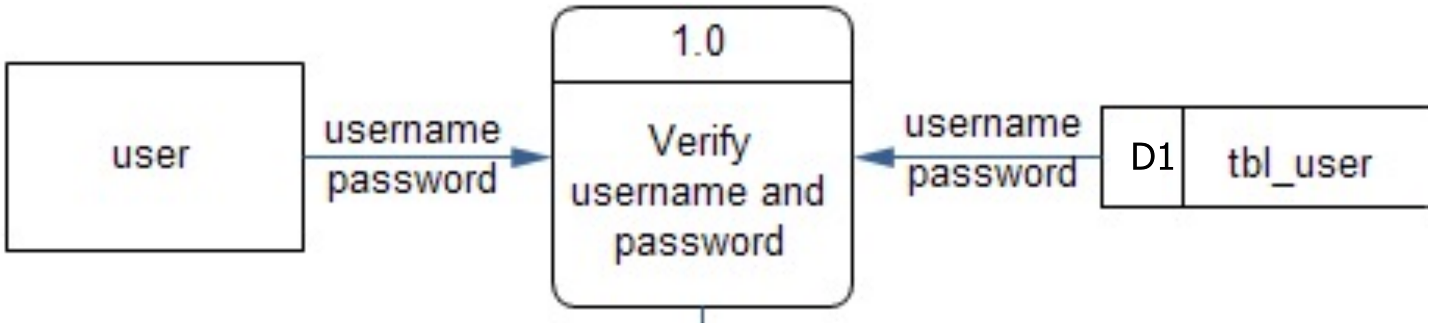


Level-0 DFD of Hoosier Burger’s food-ordering system

Meaning of a Process to Programming

- If converting a ***process*** to programming, then a process usually means **a function.**

  one process = one function / one class



Java Code:

```java
boolean verifyUser(String username, String password)
{
    //Code: Select username and password from tbl_user
    //Code: Compare them with arguments, using 'if'
    return result;
}
```

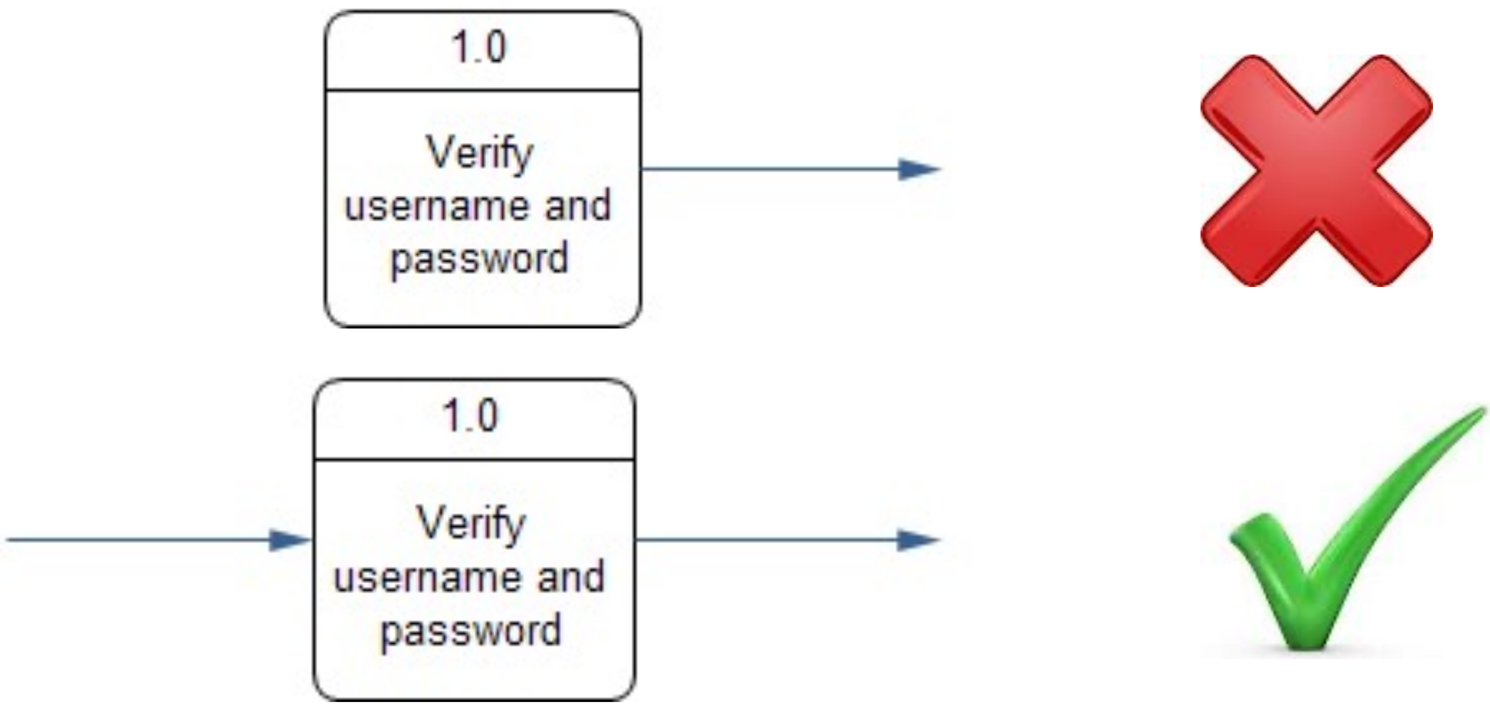
Data Flow Diagramming Rules

- There are 2 DFD guidelines that apply:
  - *The inputs to a process are different from the outputs of that process.*
    - Processes purpose is to transform inputs into outputs.
  - *Objects on a DFD have unique names.*
    - Every process has a unique name.

Data Flow Diagramming Rules (Cont.)

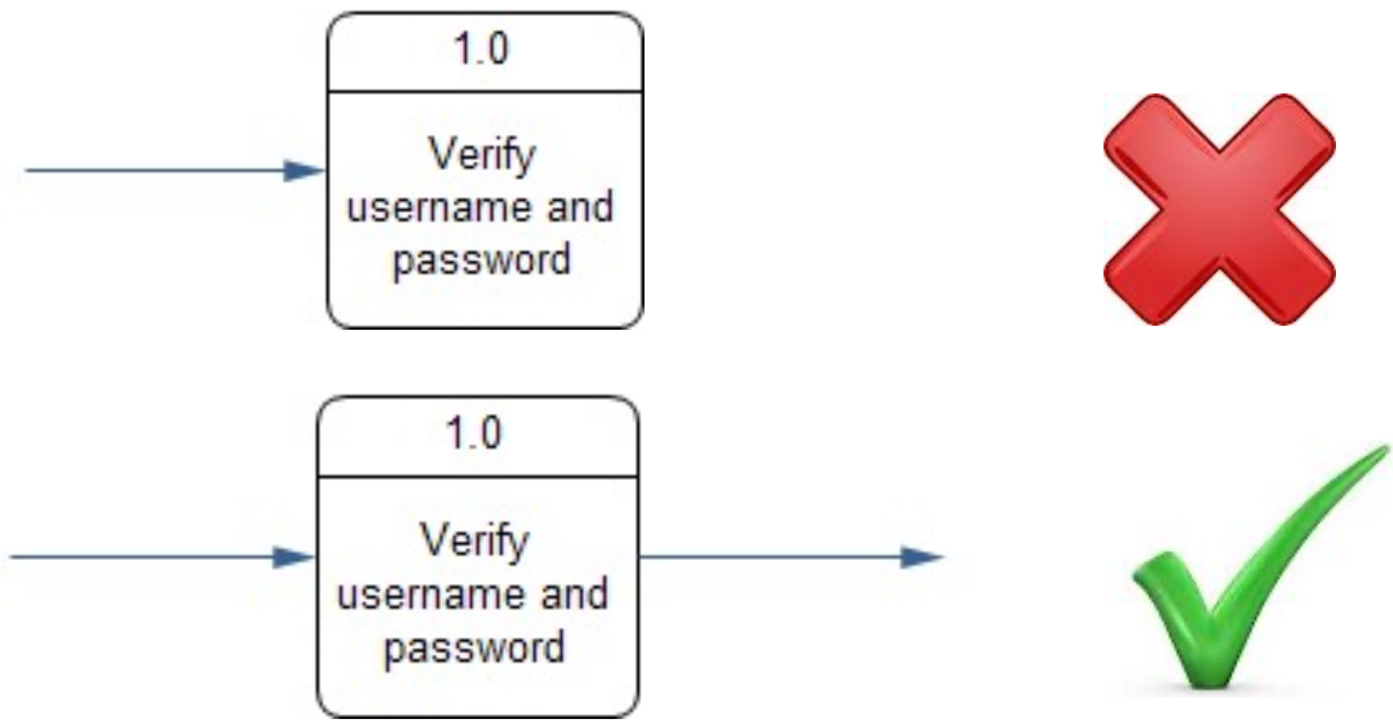
Process:

1. No process can have only output. If an object has only output, then it must a source.



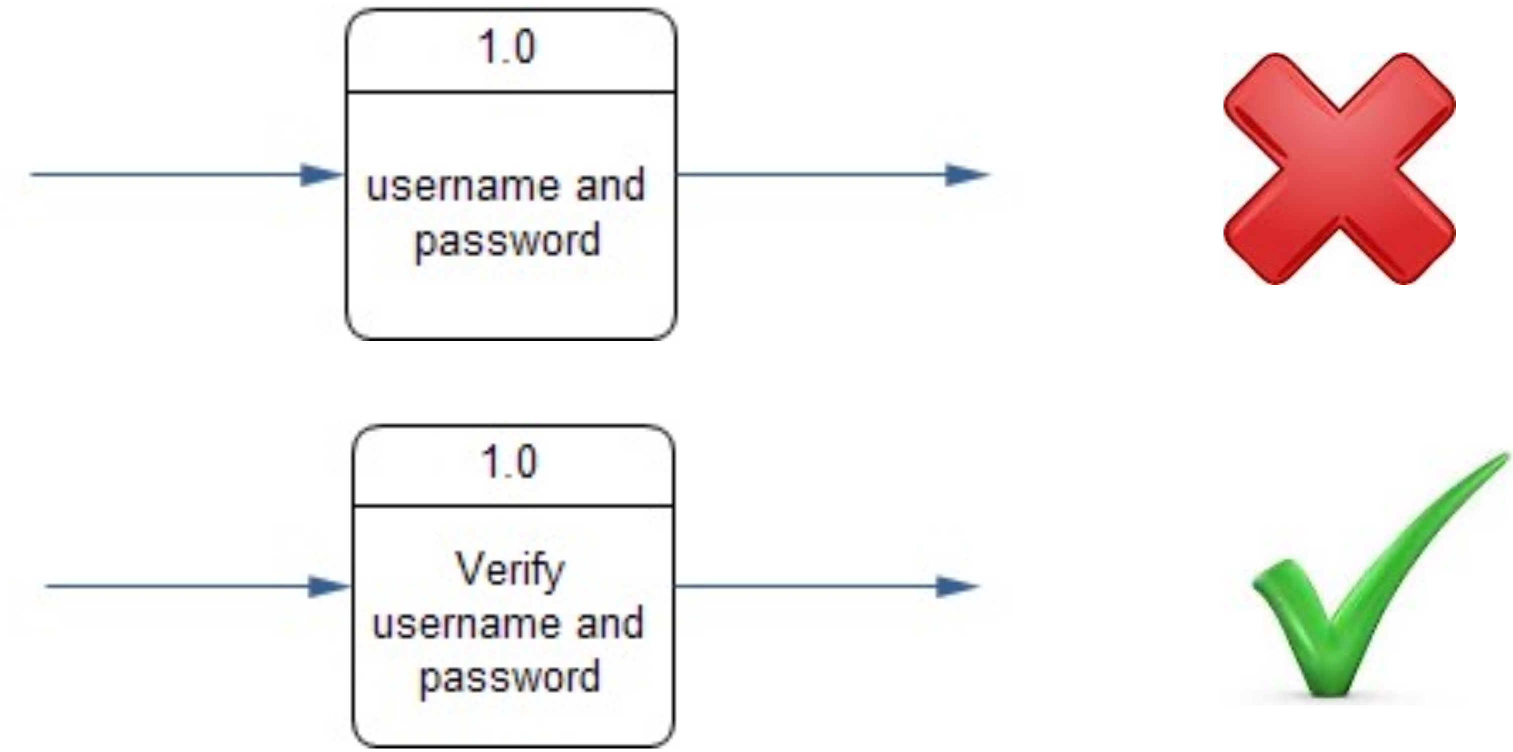
Process (cont.):

2. No process can have only input. If an object has only inputs, then it must be a sink.



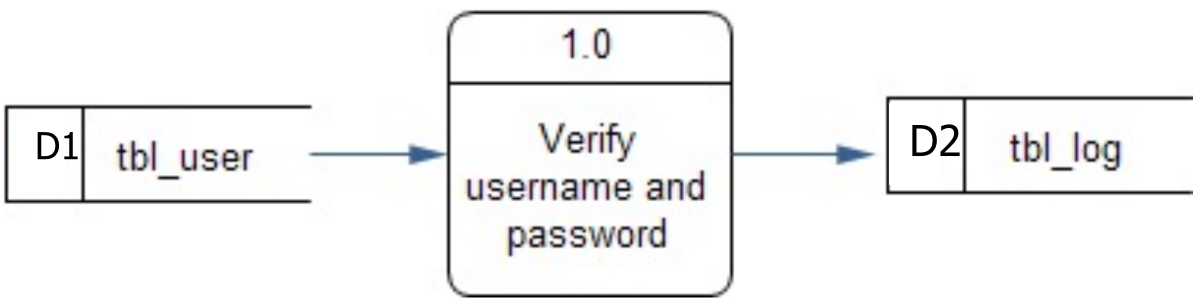
Process (cont.):

3. A process has a verb phrase label.



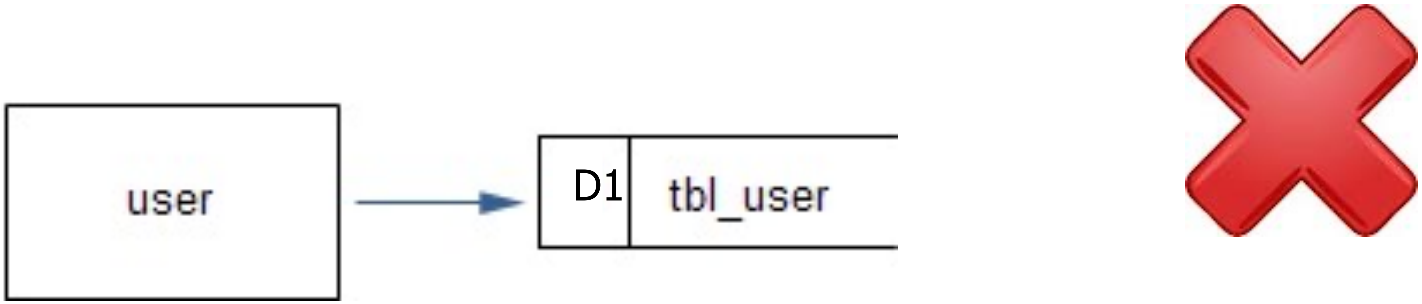
Data Store:

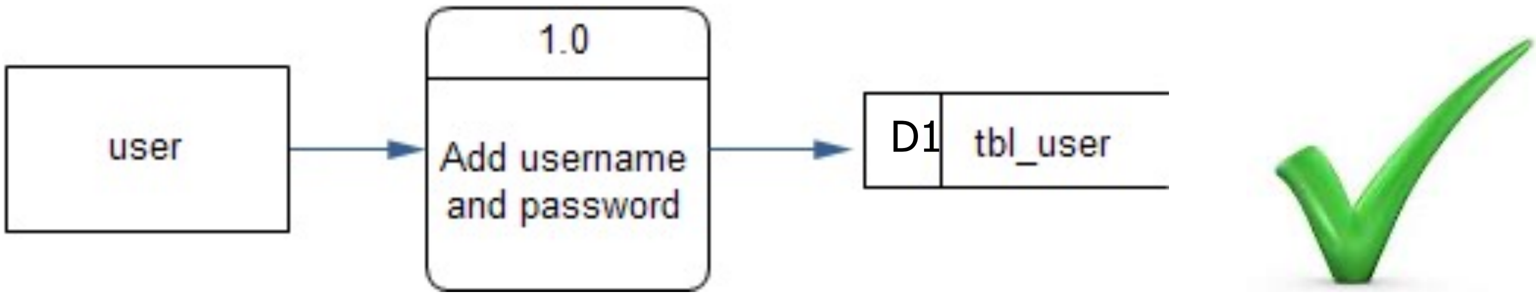
1. A data store has a noun phrase label
2. Data cannot move directly from one data store and another data store. Data must move a by process.



Data Store (cont.):

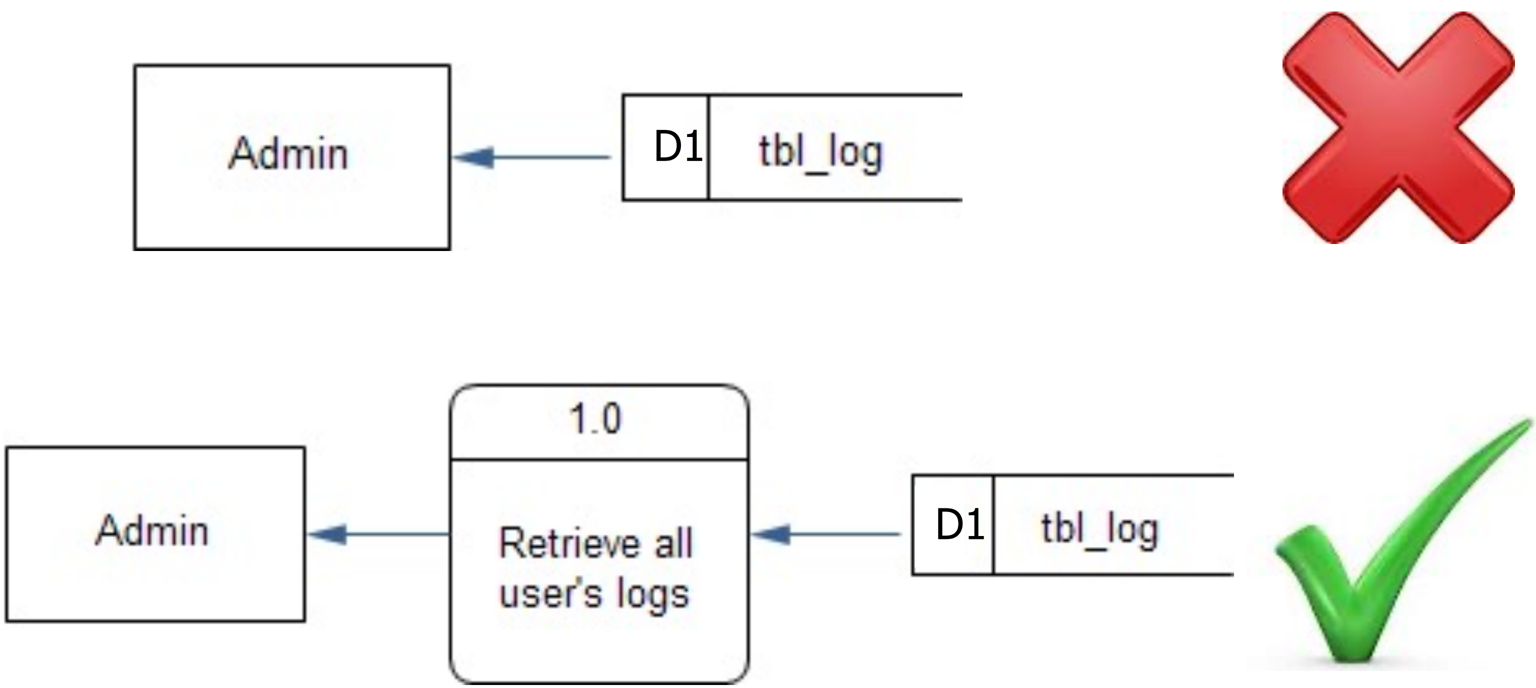
3. Data cannot move directly from an outside source to a data store. Data must move a by process.





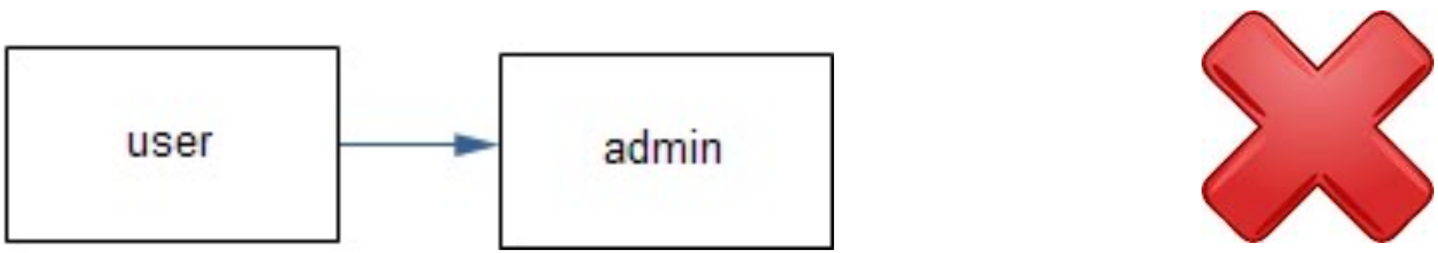
Data Store (cont.):

4. Data cannot move directly from a data store to an outside sink. Data must move a by process.



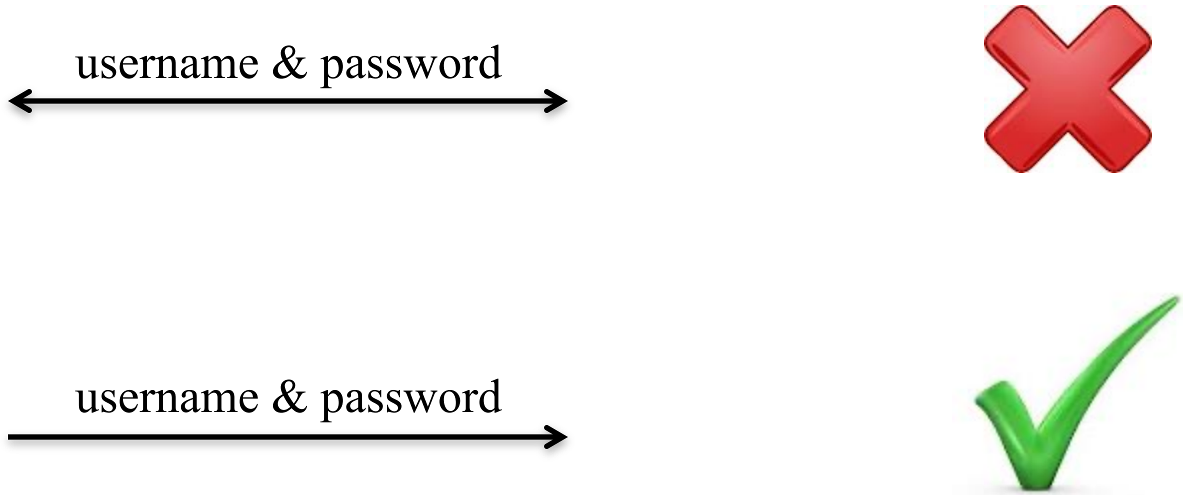
Source/Sink:

1. A source/sink has a noun phrase label.
2. Data cannot move directly from a source to a sink. Data must be moved by a process if the data are of any concern to the system. Otherwise, the data flow is not shown on the DFD.



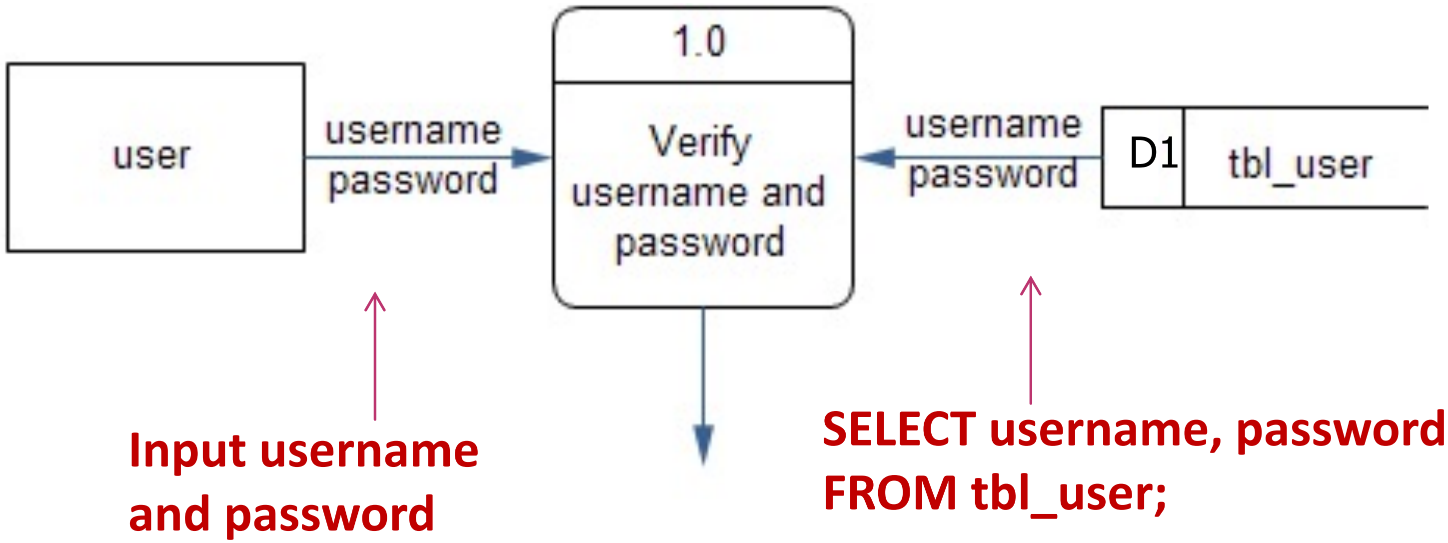
Data Flow:

1. Data flow has a noun phrase label.
2. Data flow has only one direction of flow between symbols.



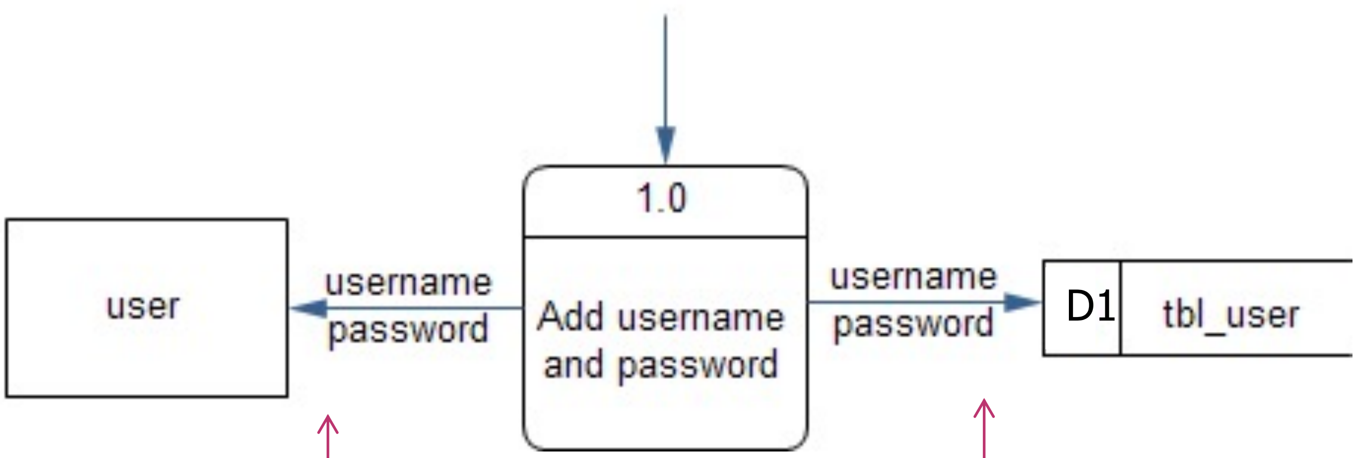
Data Flow (cont.):

3. Data flow from a source means "input" to a process.
4. Data flow from a data store means "retrieve" or "SELECT" in SQL.



Data Flow (cont.):

5. Data flow to a sink means "show"
6. Data flow to a data store can mean "update", "insert" or "delete"



Show username and password to User

INSERT INTO tbl_user VALUE (…,…,…)

Decomposition of DFDs

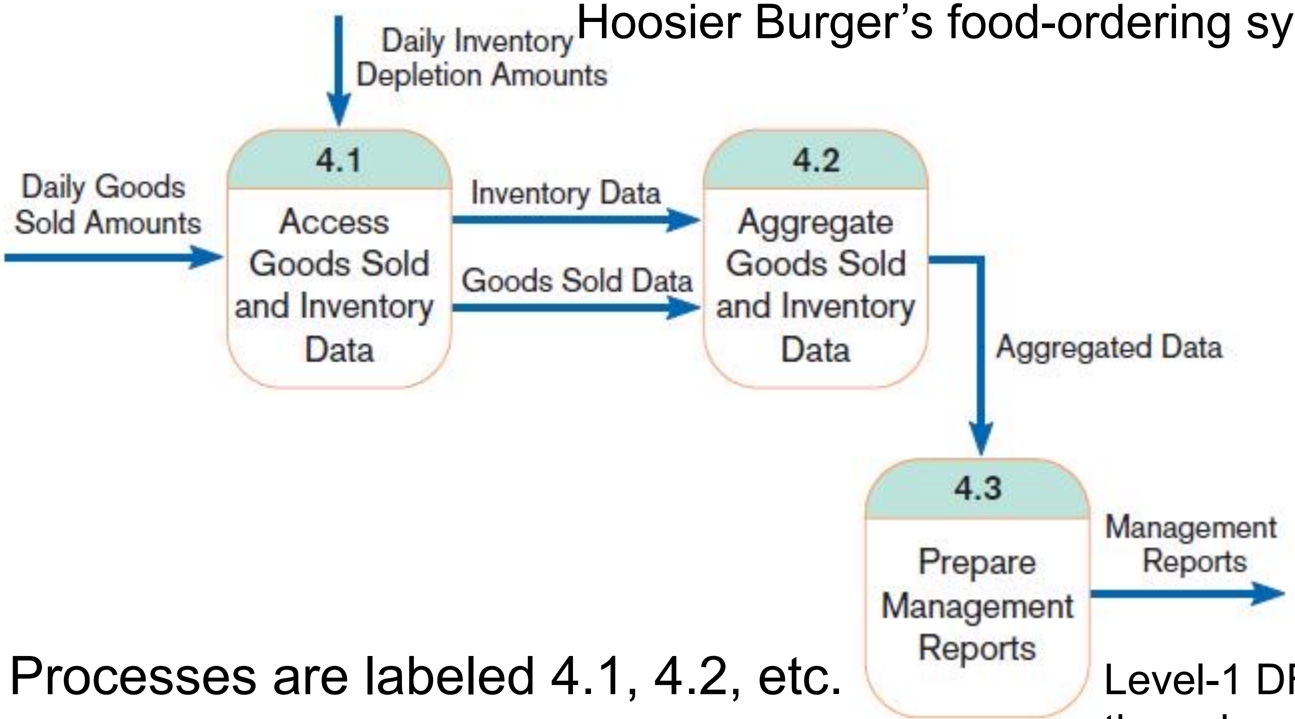
- **Functional decomposition** is an iterative process of breaking a system description down into finer and finer detail.
  - Creates a set of charts in which one process on a given chart is explained in greater detail on another chart.
  - Continues until no subprocess can logically be broken down any further.

Decomposition of DFDs (Cont.)

- *Primitive DFD* is the lowest level of a DFD.
- **Level-1 diagram** results from decomposition of Level-0 diagram.
- **Level-n diagram** is a DFD diagram that is the result of ***n*** nested decompositions from a process on a level-0 diagram.

Level-1 DFD

Level-1 diagram showing the decomposition of Process 4.0 from the level-0 diagram for Hoosier Burger's food-ordering system



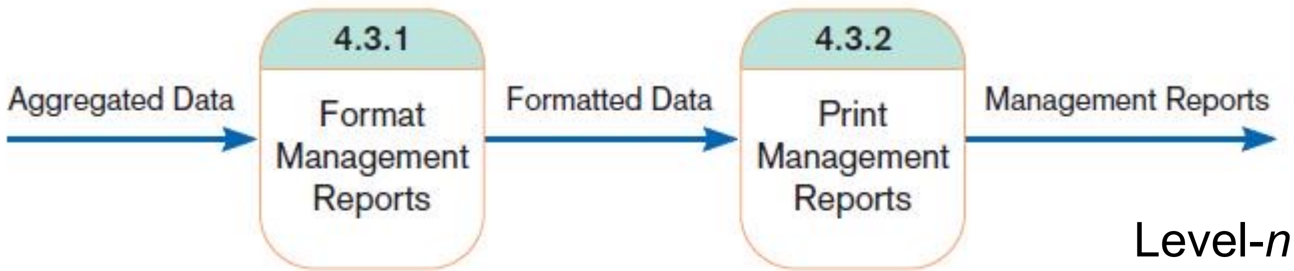
Processes are labeled 4.1, 4.2, etc. These can be further decomposed in more primitive (lower-level) DFDs if necessary.

Level-1 DFD shows the sub-processes of one of the processes in the Level-0 DFD.

This is a Level-1 DFD for Process 4.0.

Level-*n* DFD

Level-2 diagram showing the decomposition of Process 4.3 from the level-1 diagram for Process 4.0 for Hoosier Burger's food-ordering system



Level-*n* DFD shows the sub-processes of one of the processes in the Level *n-1* DFD.

This is a Level-2 DFD for Process 4.3.

Processes are labeled 4.3.1, 4.3.2, etc. If this is the lowest level of the hierarchy, it is called a *primitive DFD.*

Balancing DFDs

- **Conservation Principle**: conserve inputs and outputs to a process at the next level of decomposition
- **Balancing:** conservation of inputs and outputs to a data flow diagram process when that process is decomposed to a lower level

Balancing DFDs (Cont.)

- Balanced means:
  - Number of inputs to lower level DFD equals number of inputs to associated process of higher-level DFD
  - Number of outputs to lower level DFD equals number of outputs to associated process of higher-level DFD

Balancing DFDs (Cont.)

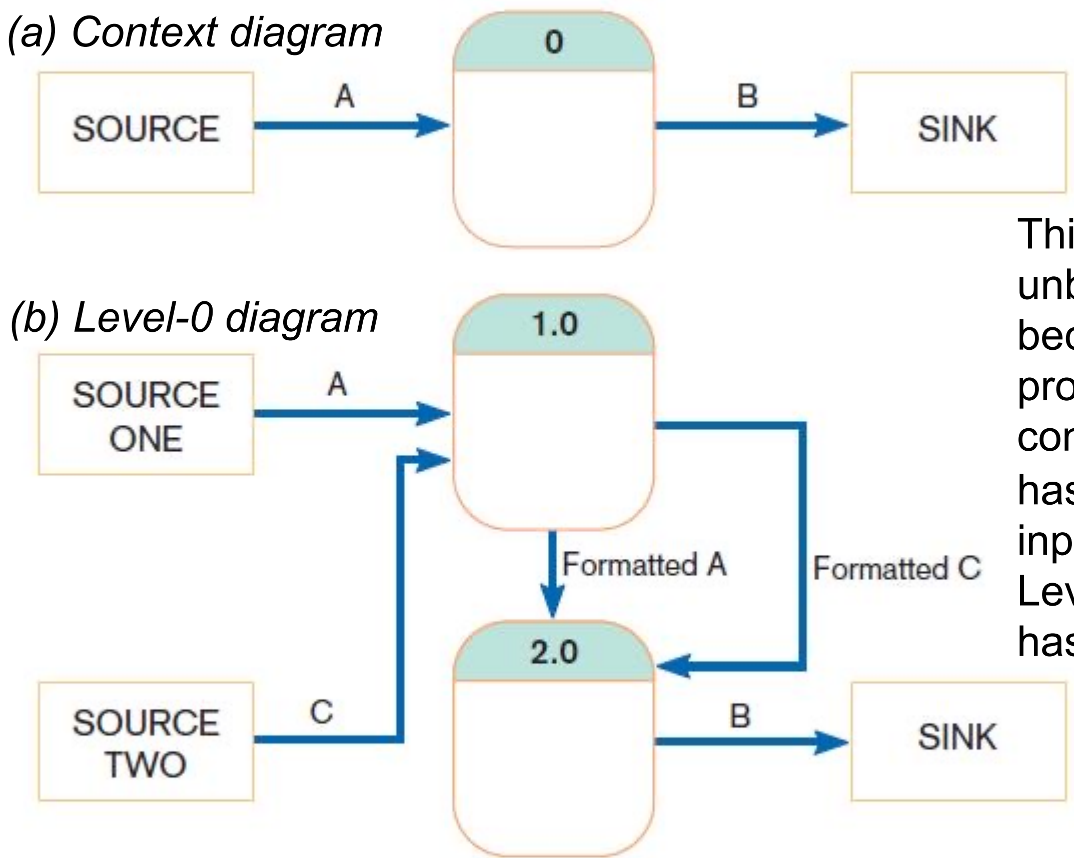
An unbalanced set of data flow diagrams

1 input

1 output

2 inputs

1 output

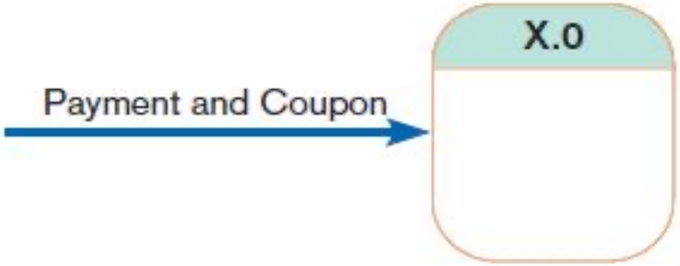


This is unbalanced because the process of the context diagram has only one input but the Level-0 diagram has two inputs.

Balancing DFDs (Cont.)

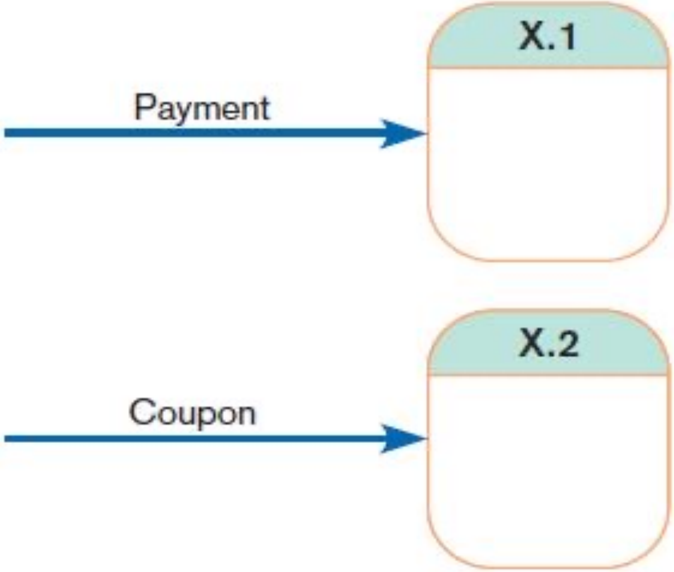
- **Data flow splitting** is when a composite data flow at a higher level is split and different parts go to different processes in the lower level DFD.
- The DFD remains balanced because the same data is involved, but split into two parts.

Balancing DFDs (Cont.)



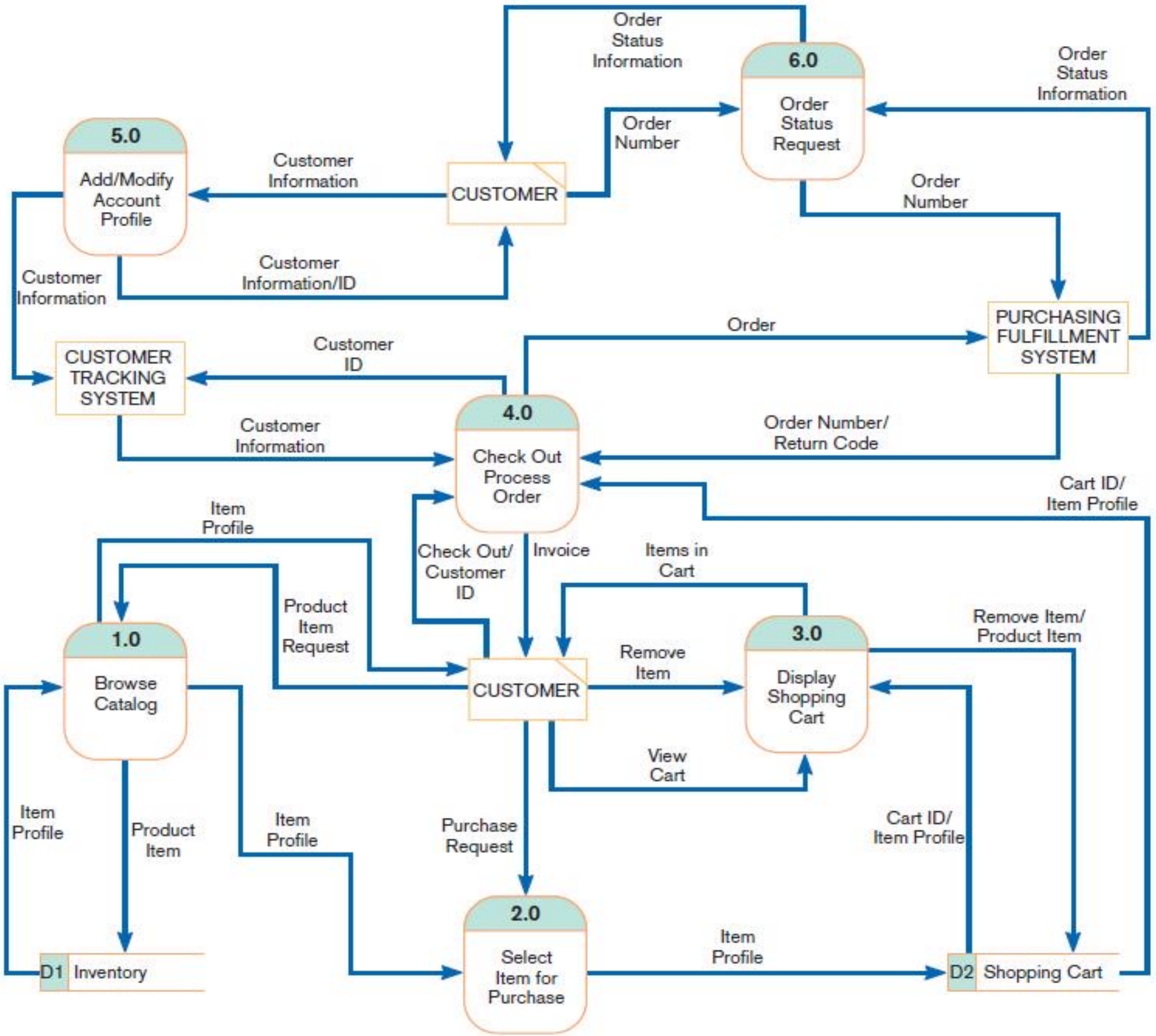
Example of data flow splitting

(a) Composite data flow



(b) Disaggregated data flows

Electronic Commerce Application: Process Modeling using Data Flow Diagrams



Level-0 data flow diagram for the WebStore